AXIOMS OF PROBABILITY

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   - Set Operations on Events

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Modeling Randomness

- Experiments that are not predictable in advance and exhibit an inherent variation in outcomes are called **random.**
- Any realistic model of a real-world phenomenon must take into account the possibility of randomness.
- A **probability model** is a mathematical representation of a random phenomenon.
  - To understand and compute the probability of an event, we need to understand the concept of the **sample space** and the **events** of an experiment.

What Is A Sample Space?

- Consider an experiment:
  - whose outcome is not predictable with certainty
  - however, the set of all possible outcomes is known.
- **Sample Space:** The set of all possible outcomes of an experiment. Usually denoted by S.

Sample Space Examples: Countable Sets

i. Experiment: A single coin flip.
   - Sample Space: $S = \{\text{head}, \text{tail}\}, \quad |S| = 2$

ii. Experiment: Two consecutive coin flips.
   - Sample Space: $S = \{HH, HT, TH, TT\}, \quad |S| = 4$

iii. Experiment: Toss of a single regular die.
   - Sample Space: $S = \{1, 2, 3, 4, 5, 6\}, \quad |S| = 6$

iv. Experiment: Toss of two dice.
   - Sample Space: $S = \{\text{all 36 possible outcomes}\}$

$$S = \left\{ \begin{array}{cccccc} (1,1), & (1,2), & (1,3), & (1,4), & (1,5), & (1,6), \\ (2,1), & (2,2), & (2,3), & (2,4), & (2,5), & (2,6), \\ (3,1), & (3,2), & (3,3), & (3,4), & (3,5), & (3,6), \\ (4,1), & (4,2), & (4,3), & (4,4), & (4,5), & (4,6), \\ (5,1), & (5,2), & (5,3), & (5,4), & (5,5), & (5,6), \\ (6,1), & (6,2), & (6,3), & (6,4), & (6,5), & (6,6) \end{array} \right\}$$

Sample Space Examples: Uncountable Sets

v. Experiment: Lifetime (hours) of a newly bought light bulb.
   - Sample Space: $S = \{x : 0 \leq x < \infty\}$
   - The possible values are uncountably many.

vi. Experiment: Water purity percentage.
   - Sample Space: $S = \{x : 0 \leq x \leq 100\}$
   - The possible values are uncountably many.

What Is An Event?

- **Event:** Any subset E of the sample space S is an event.
  - A set that consists of possible outcomes of the experiment.
  - If any of these outcomes occurs, we say that "event E occurs".

Event Examples: Countable Sample Space

i. Experiment: Two consecutive coin flips.

- Sample Space: $S = \{HH, HT, TH, TT\}$
  - Event E: Getting at least one head in two coin flips.
  - $E = \{HH, HT, TH\}$

ii. Experiment: Toss of a single regular die.

- Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$
  - Event E: Upturned face shows an even number.
  - $E = \{2, 4, 6\}$
  - Event F: Upturned face shows a greater value than 4.
  - $F = \{5, 6\}$
  - Event G: Upturned face shows neither 1 nor 3.
  - $G = \{2, 4, 5, 6\}$
- Notice that $G = E \cup F$

Event Examples: Uncountable Sample Space

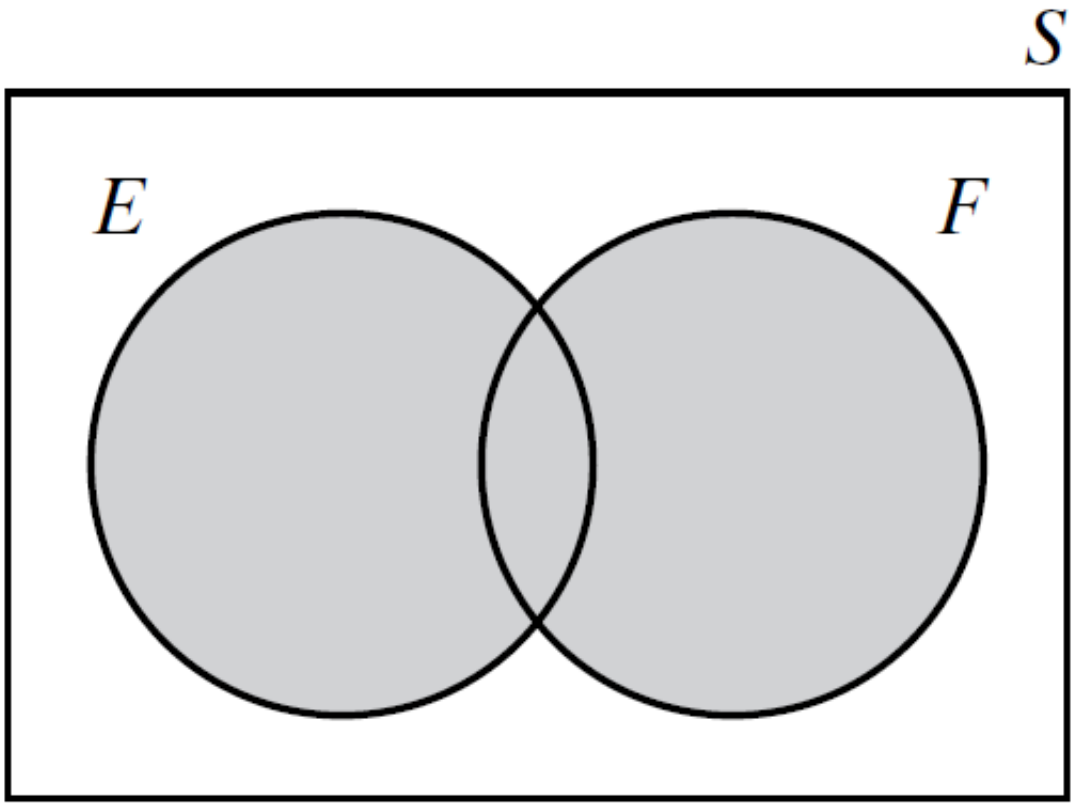
iii. Experiment: Lifetime (hours) of a newly bought light bulb.

- Sample Space: $S = \{x : 0 \leq x < \infty\}$
  - Event E: Light bulb working for at least 1000 hours.
  - $E = \{x : x \geq 1000\}$
  - Event F: Light bulb burns out in 1500 hours.
  - $F = \{x : 0 \leq x \leq 1500\}$
  - Event G: Light bulb burns out in between the 1000 and 1500 hours.
  - $G = \{x : 1000 \leq x \leq 1500\}$
- Notice that $G = E \cap F$

Union Event

- **Union Event:** For any two events E and F of a sample space S, we define the new event $E \cup F$ to consist of all outcomes that are either in E or in F or in both E and F.
- Experiment: Two consecutive coin flips.
- Sample Space: $S = \{HH, HT, TH, TT\}$
  - Event E: Getting the first flip head.
    - $E = \{HH, HT\}$
  - Event F: Getting different results in two flips.
    - $F = \{HT, TH\}$
  - Event $E \cup F$: Getting at least one head.
    - $E \cup F = \{HH, HT, TH\}$
- When $E \cup F$ occurs, we say E **or** F occurs.

Venn Representation of the Union Event

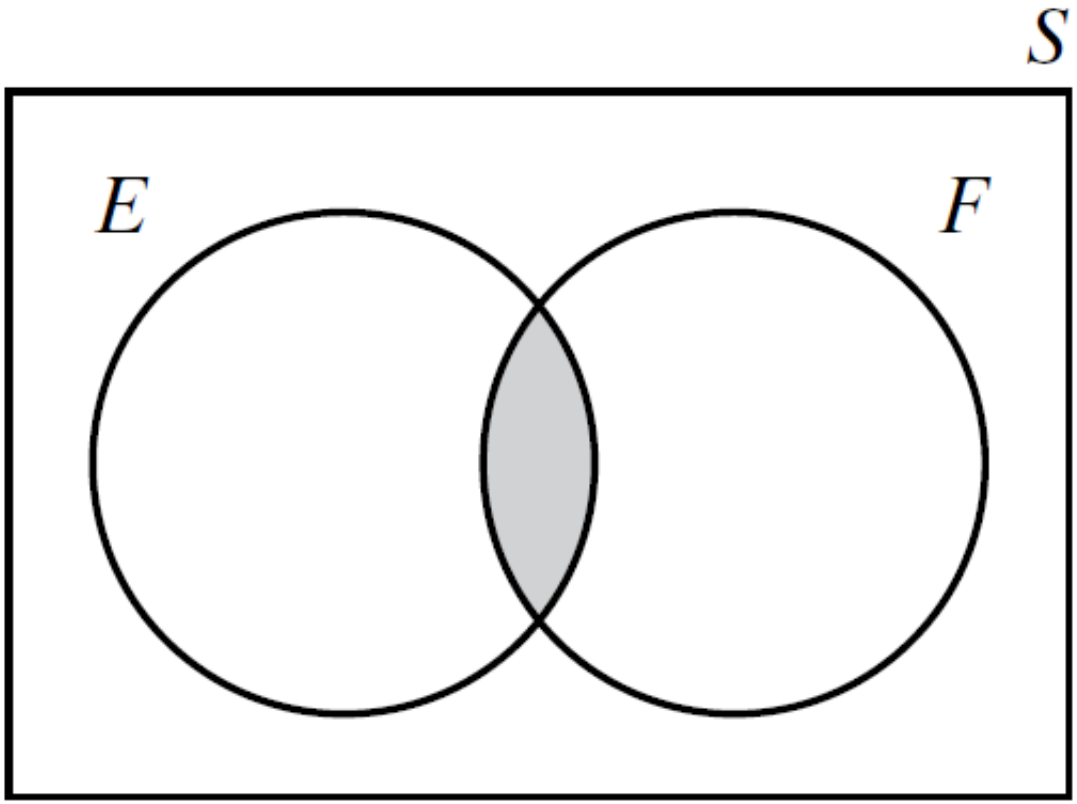


Shaded region: $E \cup F$

Intersection Event

- **Intersection Event:** For any two events E and F of a sample space S, we define the new event $E \cap F$ to consist of all outcomes that are both in E and in F.
- Experiment: Two consecutive coin flips.
- Sample Space: $S = \{HH, HT, TH, TT\}$
  - Event E: Getting the second flip tail.
    - $E = \{TT, HT\}$
  - Event F: Getting the same result in two flips.
    - $F = \{HH, TT\}$
  - Event $E \cap F$: Getting two tails.
    - $E \cap F = \{TT\}$
- When $E \cap F$ occurs, we say **both E and F** occur.

Venn Representation of the Intersection Event

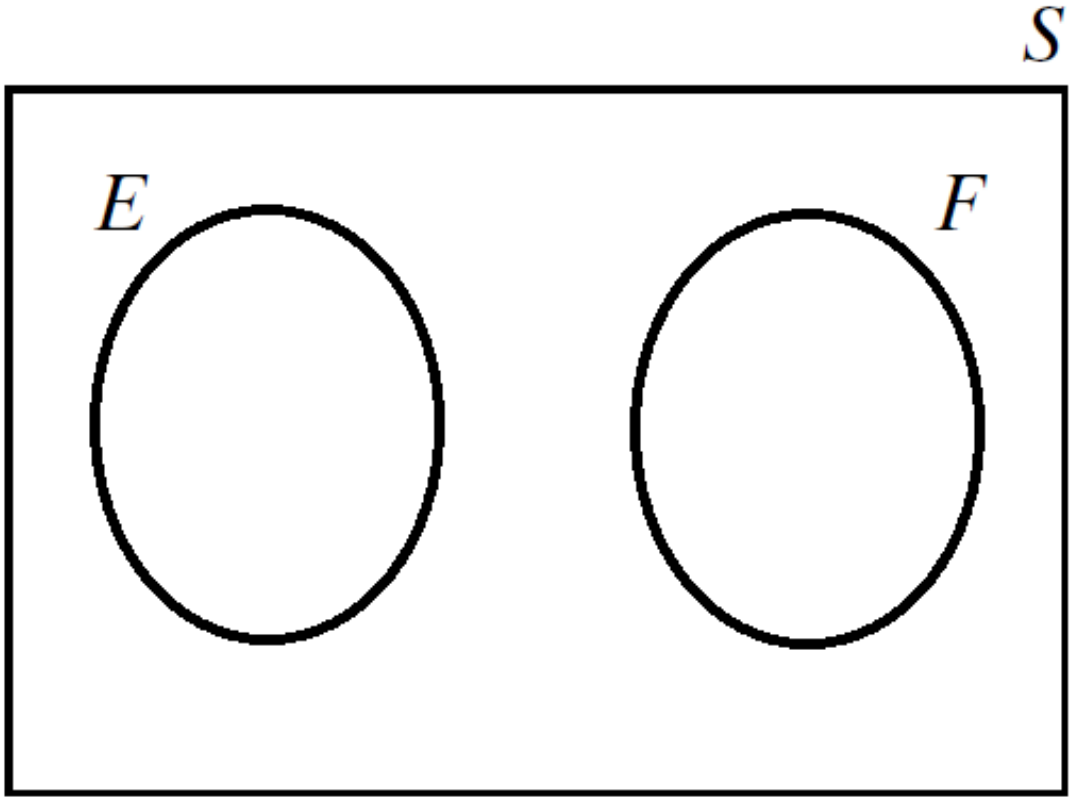


Shaded region: $E \cap F$

Mutually Exclusive (Disjoint) Events

- Two events E and F of a sample space S are said to be **mutually exclusive (disjoint)** if and only if the intersection of these events is empty.
- $E \cap F = \emptyset \Leftrightarrow E$ and F are mutually exclusive.
- Experiment: Two consecutive coin flips.
- Sample Space: $S = \{HH, HT, TH, TT\}$
  - Event E: Getting the same result in two flips.
    - $E = \{TT, HH\}$
  - Event F: Getting different results in two flips.
    - $F = \{HT, TH\}$
  - E and F are mutually exclusive ($E \cap F = \emptyset$)
- When $E \cap F = \emptyset$, we say E and F **can not occur at the same time.**

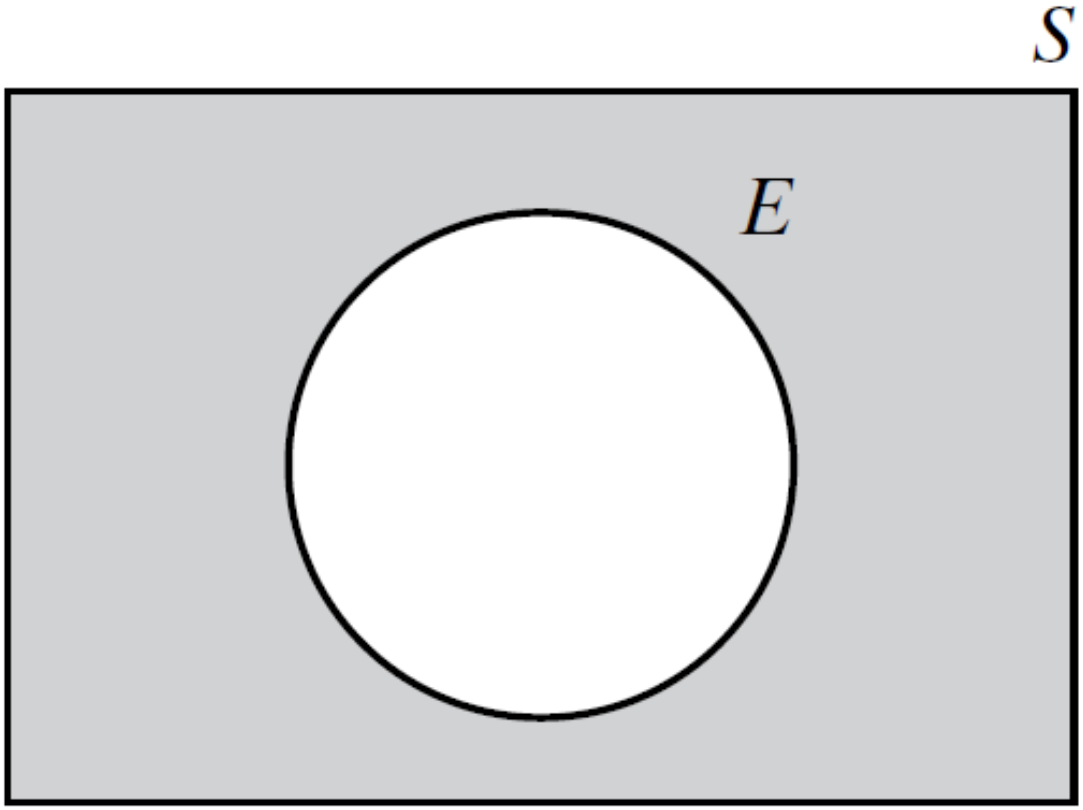
Venn Representation of Mutually Exclusive Events



Complement Event

- **Complement Event:** For any event E of a sample space S, we define E^c as the complement event and it consists of all outcomes in the sample space S that are not in E.
- Experiment: Two consecutive coin flips.
- Sample Space: $S = \{HH, HT, TH, TT\}$
  - Event E: Getting tails in both flips.
    - $E = \{TT\}$
  - Event E^c: Getting at least one heads.
    - $E^c = \{HT, TH, HH\}$
- Notice that:
  - $E \cup E^c = S$ (we say **either** E **or** E^c occurs.)
  - $E \cap E^c = \emptyset$ (complement events are mutually exclusive)

Venn Representation of the Complement Event

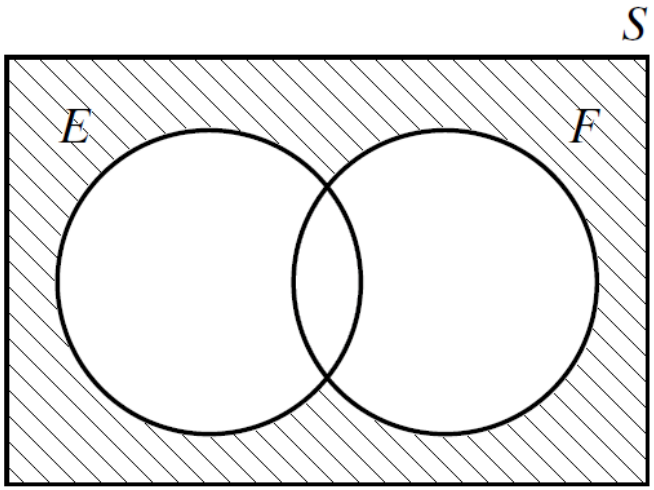


Shaded region: E^c

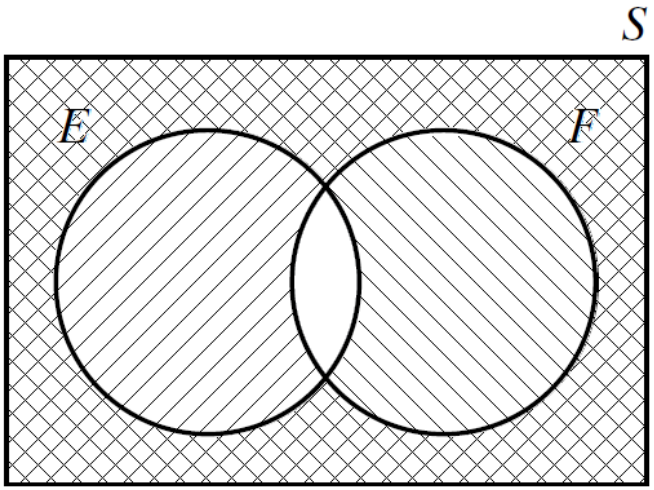
De Morgan's Laws

- **De Morgan's Laws**[1]**:** The following identities show essence in set theory, and also in probability theory:
- **Law 1:** The complement of the union of two sets is the same as the intersection of their complements:

$$(E \cup F)^c = E^c \cap F^c$$



$(E \cup F)^c$



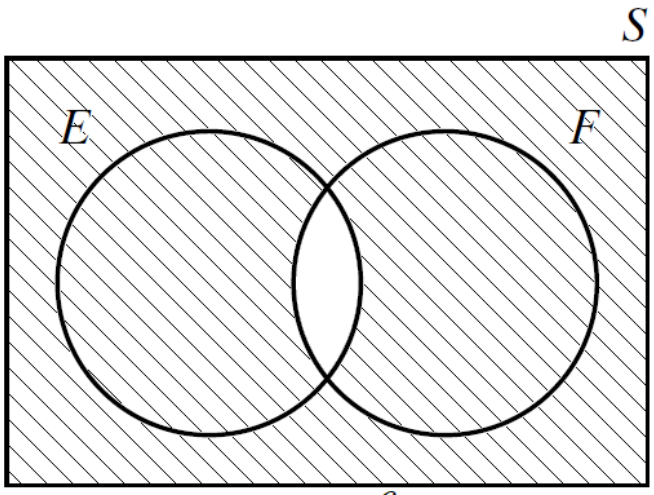
$E^c \cap F^c$

[1] Augustus De Morgan, 1806-1871.

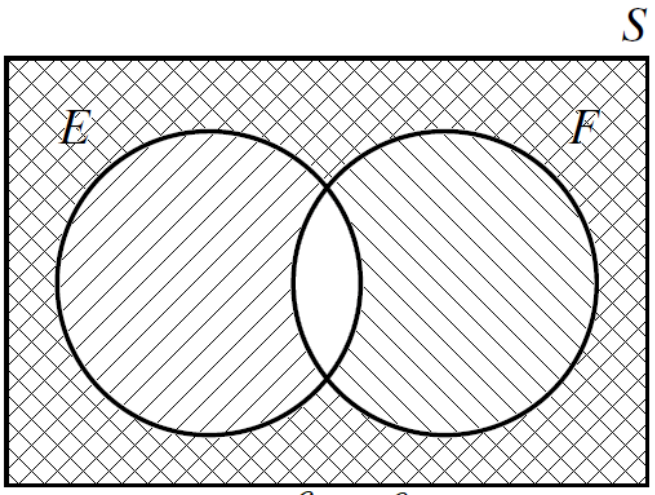
De Morgan's Laws (cont'd)

- **Law 2:** The complement of the intersection of two sets is the same as the union of their complements:

$$(E \cap F)^c = E^c \cup F^c$$



$(E \cap F)^c$



$E^c \cup F^c$

Set Operations on Events: Example

Q1. Define the sample space as the lifetime of a digital radio as:

$$S = \{x : 0 \leq x < \infty\}$$

Let A be the event that the lifetime of the radio is at least 4 years

$$A = \{x : x \geq 4\}$$

B be the event that the lifetime of the radio can not exceed 7 years

$$B = \{x : 0 \leq x \leq 7\}$$

and C be the event that the lifetime of the radio is at least 5 and at most 10 years:

$$C = \{x : 5 \leq x \leq 10\}$$

Set Operations on Events: Example (cont'd)

Q1. Use set notation to represent the following events. Provide verbal explanation as well:

$$S = \{x : 0 \leq x < \infty\} \qquad A = \{x : x \geq 4\}$$
$$B = \{x : 0 \leq x \leq 7\} \qquad C = \{x : 5 \leq x \leq 10\}$$

a. $A \cap B$
b. $A \cap C$
c. $A \cap B \cap C$
d. $A \cup C$
e. $A \cup B \cup C$
f. $A^c \cap B$
g. $B^c \cap C^c$

Set Operations on Events: Example (cont'd)

A1. The answers are as follows:

$$S = \{x : 0 \leq x < \infty\} \qquad A = \{x : x \geq 4\}$$
$$B = \{x : 0 \leq x \leq 7\} \qquad C = \{x : 5 \leq x \leq 10\}$$

a. $A \cap B = \{x : 4 \leq x \leq 7\}$
The event that the lifetime of the radio is between 4 and 7 years.

b. $A \cap C = \{x : 5 \leq x \leq 10\} = C$
The event that the lifetime of the radio is between 5 and 10 years.

c. $A \cap B \cap C = \{x : 5 \leq x \leq 7\}$
The event that the lifetime of the radio is between 5 and 7 years.

d. $A \cup C = \{x : x \geq 4\} = A$
The event that the lifetime of the radio exceeds 4 years.

Set Operations on Events: Example (cont'd)

A1. The answers are as follows:

$$S = \{x : 0 \leq x < \infty\} \qquad A = \{x : x \geq 4\}$$
$$B = \{x : 0 \leq x \leq 7\} \qquad C = \{x : 5 \leq x \leq 10\}$$

e. $A \cup B \cup C = \{x : 0 \leq x < \infty\} = S$
The whole sample space.

f. $A^c \cap B = \{x : 0 \leq x < 4\}$
The event that the lifetime of the radio is less than 4 years.

g. By De Morgan's Law, we have $B^c \cap C^c = (B \cup C)^c$.

$$B \cup C = \{x : 0 \leq x \leq 10\}$$

Then:

$$B^c \cap C^c = (B \cup C)^c = \{x : x > 10\}$$

The event that the lifetime of the radio exceeds 10 years.

Frequentists' Perspective on Probability

- Suppose an experiment (with sample space S) is repeatedly performed under exactly the same conditions.
- Let $n(E)$ denote the the number of times in the first n repetitions of the experiment that the event E occurs.
- Then the **probability** of the event E, is defined as:

$$P(E) = \lim_{n \to \infty} \frac{n(E)}{n}$$

the limiting proportion of time that E occurs.

Frequentists' Perspective on Probability (cont'd)

- Experiment: Toss of a single regular die.
- Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$
- Event E: Upturned face shows an even number.
  - $E = \{2, 4, 6\}$
- Repetitions ($n = 10000$): $3, 5, 1, 3, 6, 2, 1, 4, 2, 6, \ldots, 5, 4$
  - Say, for example, we have observed an even number 5026 times.
  - In other words, $n(E) = 5026$.
- $P(E) \approx \frac{n(E)}{n} = \frac{5026}{10000} = 0.5026$
- $P(E) = \lim_{n \to \infty} \frac{n(E)}{n} = \frac{1}{2}$

The Probability Function

- The probability is a function from the subsets of the sample space (events) to the **unit interval**, $[0, 1]$.
- The probability of an event quantifies its realization likeliness.
  - If $P(E) = 0$, event E will never occur.
  - If $P(E) = 1$, event E will definitely occur.
  - If $0 < P(E) < 1$, event E may or may not occur.
  - The possibility of occurrence is greater when $P(E)$ is large.

Axioms of Probability

There are three axioms of probability.

- **Axiom I:** Non-negativity.
- **Axiom II:** Unitarity.
- **Axiom III:** Additivity.

Axiom I: Non-negativity

- **Axiom I (Non-negativity):** The probability of an event is a non-negative value, $P(E) \geq 0$.
- **By accepting Axiom I:**
  - We can not talk about negative probabilities.
  - The probability of an event can either have zero or positive value.

Axiom II: Unitarity

- **Axiom II (Unitarity):** The probability of the sample space is always one, $P(S) = 1$.
- **By accepting Axiom II:**
  - One of the outcomes from the sample space will definitely occur.
  - Because sample space contains all events, the probability of an event can be 1 at maximum.
  - The probability of an event can only take values in $[0, 1]$, the unit interval.

$$0 \leq P(E) \leq 1$$

Axiom III: Additivity

- **Axiom III (Additivity):** Suppose E and F are any mutually exclusive (disjoint) events in the sample space ($E \cap F = \emptyset$). Then:

$$P(E \cup F) = P(E) + P(F)$$

  That is, the probability of at least one of these disjoint events occurring is the sum of their respective probabilities.
- **By accepting Axiom III:**
  - The probability of the union of **disjoint events** is the sum of the individual probabilities.
  - The probability of empty set is zero

$$P(\emptyset) = 0$$

  - For any two events A and B in the sample space, if $A \subseteq B$, then $P(A) \leq P(B)$.

Simple Propositions of Probability

By accepting the axioms of probability, we can come up with the following immediate results.

- **Proposition 1:** The probability of the complement event.
- **Proposition 2:** The probability of the union of two events.
- **Proposition 3:** The probability of the union of three events.

The Probability of the Complement Event

- **Proposition 1:** The probability of the complement event can be calculated as:

$$P(E^c) = 1 - P(E)$$

- **Proof:** Naturally, E and E^c are disjoint events. Therefore, by the third axiom:

$$P(E \cup E^c) = P(E) + P(E^c)$$

We know that $E \cup E^c = S$. It follows from the second axiom that:

$$P(E \cup E^c) = P(S) = 1$$

Thus:

$$P(E) + P(E^c) = 1 \Rightarrow P(E^c) = 1 - P(E)$$

The Probability of the Union of the Two Events

- **Proposition 2:** For any two events E and F in the same sample space:

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

- **Proof:** Left as an exercise.

The Probability of the Union of the Two Events: Example

Q2. Jane is taking two books along on her holiday vacation. With probability 0.5, she will like the first book; with probability 0.4, she will like the second book; and with probability 0.3, she will like both books. What is the probability that she likes neither book?

A2. Let:

- B_1 denote the event that Jane likes the first book,
- B_2 denote the event that Jane likes the second book.

We have the following information given by the question:

$$P(B_1) = 0.5, \quad P(B_2) = 0.4, \quad P(B_1 \cap B_2) = 0.3$$

The question asks us the following probability:

$$P((B_1 \cup B_2)^c) = ?$$

The Prob. of the Union of the Two Events: Example (cont'd)

A2. The probability that Jane likes at least one of these books is:

$$P(B_1 \cup B_2) = P(B_1) + P(B_2) - P(B_1 \cap B_2) = 0.5 + 0.4 - 0.3 = 0.6$$

The event that Jane likes neither of the books is the complement of the event that Jane likes at least one of these books. Therefore:

$$P((B_1 \cup B_2)^c) = 1 - P(B_1 \cup B_2) = 1 - 0.6 = 0.4$$

with 0.4 probability, she likes neither of the books.

The Probability of the Union of the Three Events

- **Proposition 3:** For any three events E, F, and G in the same sample space:

$$\begin{aligned} P(E \cup F \cup G) = \quad & P(E) + P(F) + P(G) \\ & -P(E \cap F) - P(E \cap G) - P(F \cap G) \\ & +P(E \cap F \cap G) \end{aligned}$$

- **Proof:** Left as an exercise.

The Probability of the Union of the Three Events: Example

Q3. An elementary school is offering 3 language classes: one in English, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 29 students in the English class, 26 in the French class, and 16 in the German class. There are 12 students that are in both English and French, 4 that are in both English and German, and 6 that are in both French and German. In addition, there are 2 students taking all 3 classes.

a. If a student is chosen randomly, what is the probability that s/he is taking at least one of the language classes?
b. If a student is chosen randomly, what is the probability that s/he is not in any of the language classes?

The Prob. of the Union of the Three Events: Ex. (cont'd)

A3. Let us define the events:

- E: The randomly selected student is taking English.
- F: The randomly selected student is taking French.
- G: The randomly selected student is taking German.

The question gives us the following information:

$$P(E) = \frac{29}{100} = 0.29, \quad P(F) = \frac{26}{100} = 0.26, \quad P(G) = \frac{16}{100} = 0.16$$

$$P(E \cap F) = \frac{12}{100} = 0.12, \quad P(E \cap G) = \frac{4}{100} = 0.04, \quad P(F \cap G) = \frac{6}{100} = 0.06$$

$$P(E \cap F \cap G) = \frac{2}{100} = 0.02$$

The Prob. of the Union of the Three Events: Ex. (cont'd)

a. The probability that the randomly selected student is taking at least one of the language classes is $P(E \cup F \cup G)$.

$$\begin{aligned} P(E \cup F \cup G) &= P(E) + P(F) + P(G) \\ &\quad - P(E \cap F) - P(E \cap G) - P(F \cap G) \\ &\quad + P(E \cap F \cap G) \\ &= 0.29 + 0.26 + 0.16 - 0.12 - 0.04 - 0.06 + 0.02 = 0.51 \end{aligned}$$

b. The probability that the randomly selected student is not in any of the language classes is $P((E \cup F \cup G)^c)$

$$P((E \cup F \cup G)^c) = 1 - P(E \cup F \cup G) = 1 - 0.51 = 0.49$$

Sample Spaces With Equally Likely Outcomes

- In many experiments, it is natural to assume that all outcomes in the sample space are equally likely to occur.
  - Consider an experiment whose sample space S is a finite and countable set, say:
    $$S = \{1, 2, \ldots, N\}$$
  - Then it is often natural to assume that
    $$P(\{1\}) = P(\{2\}) = \ldots = P(\{N\})$$
    which implies from the axioms of probability that:
    $$P(\{i\}) = \tfrac{1}{N}, \quad i = 1, \ldots, N$$
  - Then for any event E in the sample space S, the probability is[2]:
    $$P(E) = \frac{\#\,\text{of outcomes in}\,E}{\#\,\text{of outcomes in}\,S} = \frac{|E|}{|S|}$$

[2] $|E|$: Cardinality of (number of elements in) set E.

Sample Spaces With Equally Likely Outcomes (cont'd)

- In words, if we assume that all outcomes of an experiment are equally likely[3] to occur, then the probability of any event E equals the proportion of outcomes in the sample space that are contained in E.

[3] With equal probabilities.

Sample Spaces With Equally Likely Outcomes: Example

Q4. Suppose two fair dice are rolled.

a. What is the probability that the sum of the upturned faces will equal to 6?
b. What is the probability that the minimum of the upturned faces will equal to 2?
c. What is the probability that the product of the upturned faces will equal to 12?

Sample Spaces With Eq. Likely Outcomes: Ex. (cont'd)

A4. When two fair dice are rolled, the sample space consists of 36 equally likely outcomes.

$$S = \left\{ \begin{array}{llllll} (1,1), & (1,2), & (1,3), & (1,4), & (1,5), & (1,6), \\ (2,1), & (2,2), & (2,3), & (2,4), & (2,5), & (2,6), \\ (3,1), & (3,2), & (3,3), & (3,4), & (3,5), & (3,6), \\ (4,1), & (4,2), & (4,3), & (4,4), & (4,5), & (4,6), \\ (5,1), & (5,2), & (5,3), & (5,4), & (5,5), & (5,6), \\ (6,1), & (6,2), & (6,3) & (6,4), & (6,5), & (6,6) \end{array} \right\}$$

$$|S| = 36$$

Sample Spaces With Eq. Likely Outcomes: Ex. (cont'd)

a. E: The sum of the upturned faces will equal to 6.

$$E = \{(1,5),(2,4),(3,3),(4,2),(5,1)\}$$

$$S = \left\{ \begin{array}{llllll} (1,1), & (1,2), & (1,3), & (1,4), & (1,5)^*, & (1,6), \\ (2,1), & (2,2), & (2,3), & (2,4)^*, & (2,5), & (2,6), \\ (3,1), & (3,2), & (3,3)^*, & (3,4), & (3,5), & (3,6), \\ (4,1), & (4,2)^*, & (4,3), & (4,4), & (4,5), & (4,6), \\ (5,1)^*, & (5,2), & (5,3), & (5,4), & (5,5), & (5,6), \\ (6,1), & (6,2), & (6,3) & (6,4), & (6,5), & (6,6) \end{array} \right\}$$

$$P(E) = \frac{|E|}{|S|} = \frac{5}{36}$$

Sample Spaces With Eq. Likely Outcomes: Ex. (cont'd)

b. F: The minimum of the upturned faces will equal to 2.

$$F = \{(2,2),(2,3),(2,4),(2,5),(2,6),(3,2),(4,2),(5,2),(6,2)\}$$

$$S = \left\{ \begin{array}{llllll} (1,1), & (1,2), & (1,3), & (1,4), & (1,5), & (1,6), \\ (2,1), & (2,2)^*, & (2,3)^*, & (2,4)^*, & (2,5)^*, & (2,6)^*, \\ (3,1), & (3,2)^*, & (3,3), & (3,4), & (3,5), & (3,6), \\ (4,1), & (4,2)^*, & (4,3), & (4,4), & (4,5), & (4,6), \\ (5,1), & (5,2)^*, & (5,3), & (5,4), & (5,5), & (5,6), \\ (6,1), & (6,2)^*, & (6,3) & (6,4), & (6,5), & (6,6) \end{array} \right\}$$

$$P(F) = \frac{|F|}{|S|} = \frac{9}{36} = 0.25$$

Sample Spaces With Eq. Likely Outcomes: Ex. (cont'd)

c. G: The product of the upturned faces will equal to 12.

$$G = \{(2,6),(6,2),(3,4),(4,3)\}$$

$$S = \left\{ \begin{array}{cccccc} (1,1), & (1,2), & (1,3), & (1,4), & (1,5), & (1,6), \\ (2,1), & (2,2), & (2,3), & (2,4), & (2,5), & (2,6)^*, \\ (3,1), & (3,2), & (3,3), & (3,4)^*, & (3,5), & (3,6), \\ (4,1), & (4,2), & (4,3)^*, & (4,4), & (4,5), & (4,6), \\ (5,1), & (5,2), & (5,3), & (5,4), & (5,5), & (5,6), \\ (6,1), & (6,2)^*, & (6,3) & (6,4), & (6,5), & (6,6) \end{array} \right\}$$

$$P(G) = \frac{|G|}{|S|} = \frac{4}{36} = \frac{1}{9}$$

Sample Spaces With Equally Likely Outcomes: Ex. II

Q5. If 3 balls are randomly drawn from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?

Sample Spaces With Equally Likely Outcomes: Ex. II

Q5. If 3 balls are randomly drawn from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?

A5. First, we need to calculate the number of different ways of selecting 3 balls out of 11, that is the size of our sample space - number of all possible outcomes.

$$|S| = \binom{11}{3} = 165$$

Sample Spaces With Eq. Likely Outcomes: Ex. II (cont'd)

A5. On the other hand, the event of interest, E, is selecting one white and two black balls.

- We need to select 1 white ball out of 6 white balls,
- the other 2 should be black out of 5 black balls.

The number of ways of selecting these balls is:

$$|E| = \binom{6}{1}\binom{5}{2} = 6 \times 10 = 60$$

Thus, the probability of the event of interest is:

$$P(E) = \frac{|E|}{|S|} = \frac{\binom{6}{1}\binom{5}{2}}{\binom{11}{3}} = \frac{\frac{6!}{1!5!}\frac{5!}{2!3!}}{\frac{11!}{3!8!}} = \frac{6 \times 10}{165} = \frac{60}{165} = \frac{4}{11}$$

Sample Spaces With Equally Likely Outcomes: Ex. III

Q6. An urn contains 100 balls, two of which are special. If 60 of these balls are withdrawn one at a time, with each selection being equally likely to be any of the remaining balls at that time, what is the probability that **at least one** special ball is chosen in these 60 balls?

Sample Spaces With Eq. Likely Outcomes: Ex. III (cont'd)

A6. Let us, first, compute the probability of selecting none of these special balls. For that, all 60 balls should be selected among the remaining 98 balls.

$$\frac{\binom{2}{0}\binom{98}{60}}{\binom{100}{60}} = \frac{\frac{98!}{60!38!}}{\frac{100!}{60!40!}} = \frac{98!}{60!38!} \times \frac{60!40!}{100!} = \frac{26}{165}$$

The desired event, selecting at least one special ball is the complement of the above event. Therefore, the probability of selecting at least one special ball is:

$$1 - \frac{26}{165} = \frac{139}{165}$$

Sample Spaces With Equally Likely Outcomes: Ex. IV

Q7. In a class of 21 students, there are 3 project teams of 7 members. If we randomly select 6 students from the class, what is the probability of selecting 2 students from each team?

Sample Spaces With Equally Likely Outcomes: Ex. IV

Q7. In a class of 21 students, there are 3 project teams of 7 members. If we randomly select 6 students from the class, what is the probability of selecting 2 students from each team?

A7. The desired probability is:

$$\frac{\binom{7}{2}\binom{7}{2}\binom{7}{2}}{\binom{21}{6}} = \frac{\frac{7!}{2!5!}\frac{7!}{2!5!}\frac{7!}{2!5!}}{\frac{21!}{6!15!}} = \frac{441}{2584} \approx 0.1707$$

In the numerator, the generalized principle of counting is used.

Sample Spaces With Equally Likely Outcomes: Ex. V

Q8. There are 5 hotels in a certain town. If 3 guests randomly check into hotels in a day, what is the probability that they each check into a different hotel?

Sample Spaces With Equally Likely Outcomes: Ex. V

Q8. There are 5 hotels in a certain town. If 3 guests randomly check into hotels in a day, what is the probability that they each check into a different hotel?

A8. Because each guest can choose among 5 hotels with equal chances, the sample space consists of $5^3 = 125$ equally likely outcomes.

$$|S| = 5^3 = 125$$

Sample Spaces With Eq. Likely Outcomes: Ex. V (cont'd)

A8. The desired event is for guests to choose different hotels. While the first guest can check into any of the 5 hotels, for this event to occur:

- the second guest can only check into the remaining 4 hotels,
- then, the third guest can only check into the remaining 3 hotels.

Thus, the number of outcomes in the sample space that point this event is:

$$|E| = 5 \times 4 \times 3 = 60$$

The desired probability is:

$$P(E) = \frac{|E|}{|S|} = \frac{60}{125} = \frac{12}{25}$$

END OF LECTURE SLIDE 02